Detecting Text-Reading Hallucinations in Vision-Language Descriptions of Cyrillic Signage, Packaging and Documents for Blind Users

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Abstract

Blind people increasingly rely on vision-language models (VLMs) to read signs, product labels and documents aloud, but a model that cannot read a word tends to produce a plausible one instead. We measure how often this happens for Ukrainian (Cyrillic) text and whether it can be detected at run time. We built a benchmark of 567 synthetic, photograph-like images of signage, packaging and documents. Every character is annotated, together with its visibility in the frame. The design crosses image quality, framing (cropping by the frame edge, finger or glare occlusion) and the type of the distinctive name (a real word, an invented word, or a word one letter away from a real one). Four Claude models described each image for a blind user and, separately, transcribed it. Only 2.8–4.2% of the quoted words were hallucinated, yet 35–46% of the descriptions contained at least one hallucinated word, including 11–19% of descriptions of clean, fully framed images. Invented and near-word names were hallucinated at 19–38%, against 1.6–2.0% for common words. Near-words were normalised to the real word, letters specific to Ukrainian were replaced by Russian-like spellings, and text hidden by the frame was completed. Only 39–46% of hallucinated words were hedged. Combining the model's own re-reading, an independent Tesseract OCR pass and agreement between models detected hallucinated words with an AUROC of 0.95–0.97, catching 85–91% of them at a 10% false-flag rate. We derive design recommendations for assistive reading applications.

Keywords: blind and visually impaired users, vision-language models, hallucination detection, scene text reading, Cyrillic script, assistive technology

1 Introduction

Reading text is one of the most frequent reasons blind people photograph their surroundings. The first crowd-powered visual question answering service, VizWiz, already found that a large share of questions asked what a label, sign or screen said [1], and later analyses of tens of thousands of such questions confirmed that reading and product identification dominate everyday visual needs [2, 3]. Dedicated text readers such as Seeing AI and OrCam serve this need [4, 5], and image-processing methods can adapt other visual content, such as illustrations, for low-vision readers [6]. Since 2023, general-purpose vision-language models (VLMs) have increasingly taken over the task. Scene-description apps built on these models let a blind user point a phone at a medicine box or a utility bill and receive a fluent spoken description [7–9].

Fluency is the problem. A conventional OCR engine that cannot read a word usually produces visible garbage, and a sighted user would notice it. A VLM that cannot read a word tends to produce a *plausible* word instead: a common brand name in place of an unusual one, a complete street name where the sign is cut off by the edge of the frame, or a date that is one digit off. For a blind user the description is often the only access to the text. Blind users are known to place considerable trust in machine-generated descriptions even when they are wrong [10], and they have few independent ways to check them [11, 12]. A hallucinated expiry date, dosage or amount due is therefore not a cosmetic error but an accessibility failure with practical consequences.

This risk is plausibly greater for Cyrillic text, and for Ukrainian in particular. Text-centric benchmarks report clearly lower VLM accuracy for non-Latin scripts [13, 14]. Ukrainian adds letters that do not exist in Russian (і, ї, є, ґ) and an apostrophe inside words, so a model with a strong Russian or generic Cyrillic prior can silently "correct" them. Ukrainian everyday text is also dense with proper names, such as brands, surnames, street names and invented shop names, which a language prior cannot predict.

This paper asks two questions:

RQ1 When a VLM describes a photograph of Cyrillic signage, packaging or a document for a blind user, how often does it assert text that is not printed there, and what drives these text-reading hallucinations?
RQ2 Can such hallucinations be detected automatically, without ground truth, using signals that an assistive application could compute at run time?

Answering RQ1 requires exact knowledge of what is printed and of what is actually visible in the photograph. Neither is available for real photographs without expensive annotation, and visibility is not available even then. We therefore built a controlled benchmark of 567 synthetic but photograph-like images of Ukrainian signs, product labels and documents. For every image we know every character and whether it is fully visible, partly visible, or hidden by the frame edge or by an occluding finger

or glare. The design crosses three genres, three levels of image quality, three framing conditions, and three conditions for the stimulus's distinctive name: a real word, an invented word, or a *near-word* that differs from a real word in a single letter. The near-word condition directly probes whether models normalise unusual text towards what they expect to see.

We evaluated four Claude models (Haiku 5.5, Sonnet 5.5, Opus 5.5 and Fable 5.1) with a blind-user description prompt and a constrained transcription prompt. Every word the models quote as printed text was labelled automatically against the ground truth, and the labeller was checked by a manual audit. We then evaluated five run-time detectors and their combination: the model's own re-reading, agreement with other models, an independent Tesseract OCR pass, a lexicon check and verbalised hedging.

The contributions are: (i) a reproducible, fully annotated benchmark design for text-reading hallucination in assistive image description, with released generator code, stimuli and response-free results; (ii) an error taxonomy grounded in visibility, separating misreadings and fabrications of visible text from *completions* of text the camera did not capture; (iii) measurements of how image quality, framing, text category and Ukrainian-specific orthography drive these errors in current models; and (iv) an evaluation of detectors that an assistive application could deploy today, with operating points stated in terms of how many hallucinated words are caught at a given rate of false alarms on correct words.

2 Related work

2.1 AI visual assistance for blind and low-vision people

Remote sighted assistance and crowd-sourced answers [1] were followed by datasets of real photographs taken by blind people, which made their characteristics measurable: VizWiz images are frequently blurred, poorly framed or partly occluded, and many of them contain text [3, 15]. What blind and low-vision people want from image descriptions depends on the context, but transcribing visible text is consistently valued [16]. Earlier computer-vision aids for visually impaired people focused on detection and recognition pipelines [5, 17]. Text readers built on such pipelines are now an established tool, although their accuracy degrades on real-world photographs [4].

Large multimodal models changed the interaction from reading out recognised text to conversational description. Diary and interview studies of apps such as Be My AI report broad adoption and new use cases, alongside errors that users find hard to detect and strategies for verifying output, such as re-taking photographs or asking again [7–9]. Studies of how blind people handle recognition errors show that verification is laborious and often impossible without sighted help [11, 12]. Blind users also tend to trust captions that are inconsistent with the context [10]. These findings motivate detection that works *inside* the assistive pipeline rather than relying on the user.

2.2 Hallucination in vision-language models

Object hallucination in image captioning was quantified early with the CHAIR metric, which attributed it partly to language priors [18]. For instruction-tuned VLMs it has been probed with polling questions in POPE [19] and with adversarial visual questions in HallusionBench [20]. Recent surveys organise the causes into data, model and decoding factors, and highlight the dominance of the language prior when visual evidence is weak [21, 22]. We use the same intuition but measure it on text, where the visual evidence can be degraded in a controlled way. Our earlier work on inclusive image captioning with large language models [23] examined description quality for visually impaired users. The present study isolates the text-reading component of such descriptions.

2.3 Reading text with vision-language models

Scene-text benchmarks such as TextVQA [24], ST-VQA [25] and TextCaps [26] established that answering and captioning often require reading. OCRBench showed that large multimodal models rely on semantic plausibility: they read meaningful words far better than non-semantic character strings [13]. MTVQA found a substantial accuracy gap for non-Latin scripts [14]. For Ukrainian food packaging specifically, the choice of visual model measurably changes the quality of optical reading of label information [27]. Text in an image also strongly steers model behaviour, as typographic attacks show [28]. Closest to our work, recent studies document OCR hallucination by multimodal models on degraded documents [29] and fluent fabrication by VLM-based OCR on historical documents even where the character error rate is low [30]. None of these studies controls what is visible in the image or targets the assistive setting, Cyrillic script or Ukrainian orthography. We use synthetic rendering, a standard way to obtain exact text ground truth [31, 32], to provide this control.

2.4 Detecting hallucinations without ground truth

Black-box detectors exploit the observation that hallucinated content is unstable across samples: SelfCheckGPT compares a response with stochastic re-samples [33], and semantic entropy measures disagreement among sampled meanings [34, 35]. Models can also be asked to state their confidence [36, 37]. However, verbalised confidence tends to be overconfident [38], and hedging language does not faithfully reflect a model's intrinsic uncertainty [39]. For text reading, an independent OCR engine such as Tesseract [40] offers a cross-check that does not share the VLM's language prior. Our study compares these families of signals on the same quoted words.

3 Method

3.1 Stimuli

Content.

A generator written for this study produces Ukrainian text in three genres that blind users commonly need to read. *Signage* covers shop signs with opening hours, street-name plates, notices and public-transport stops. *Packaging* covers food labels with net weight, ingredients, best-before date and manufacturer, medicine boxes with dose, count, expiry date and batch number, and household chemicals with hazard warnings. *Documents* cover utility bills, shop receipts, medical referrals and building announcements. Content is assembled from curated Ukrainian vocabulary (e.g., 18 shop types, 30 real street names, 18 cities, real first names and patronymics) and random numerals (dates, times, prices, phone numbers, account and batch numbers). Each ground-truth token carries the category of the slot that produced it: *common word*, *real name*, *invented name*, *near-word name* or *digits*.

Every stimulus has one distinctive *name slot*: a brand, shop, company, street or patient surname, or a drug name. Its content is set by a balanced factor with three levels. In the *word* level it is a real word or name typical of Ukrainian brands (e.g., «Ластівка», «Барвінок») or a real surname or drug name. In the *invented* level it is a pronounceable string of Ukrainian syllables that is absent from a 290-thousand-word Ukrainian frequency list derived from OpenSubtitles[1] (e.g., «Фецедвус»). In the *near-word* level it is a real word with exactly one letter replaced by another letter of the same class, again checked to be absent from the word list (e.g., «Воложка» from «Волошка», «Метфермін» from «Метформін»). The near-word level is the sharpest probe of prior-driven normalisation: a model that reads what it expects rather than what is printed will "correct" it.

Rendering.

Lines are typeset on a panel styled for the genre: coloured sign plates with optional frames, pale product labels, or white and off-white paper, the latter with a monospaced font for receipts. Nineteen Cyrillic-capable font families are used, including serif, sans-serif, condensed, monospaced and decorative or handwritten faces; decorative faces appear only in titles. The panel is placed in a textured background scene through a random homography with small perspective distortion and rotation, so that it resembles a hand-held photograph. Because every character's box is known in panel coordinates, its position in the final image is known exactly.

Framing.

The second factor controls what reaches the camera. *Full frame* keeps the whole panel with a margin. *Crop* moves one frame edge (left, right or bottom) into the panel by 14–35%, which cuts lines and words as happens when a blind user aims a phone imprecisely [3]. *Occlusion* places either a skin-coloured finger entering from the bottom edge or a specular glare highlight over a random character of a random line. For each

[1] https://github.com/hermitdave/FrequencyWords (2018 release, CC BY-SA 4.0).



Figure 1 Example stimuli from the three genres under different quality and framing conditions. All text is Ukrainian, and the ground truth records every character and whether it is visible in the final frame.

character we test a 3×3 grid of points inside its box. A character is *visible* if at least 8 of the 9 points are inside the frame and not occluded, *hidden* if at most 2 are, and *partial* otherwise. For glare, a point counts as occluded where the highlight's opacity exceeds 0.85. A token's state follows from its characters: visible, hidden, or partial if its characters are mixed. Crop and occlusion stimuli were re-drawn until at least one token was not fully visible, and full-frame stimuli until none was hidden.

Quality.

The third factor degrades the image after it is scaled to 1024 pixels on the long side. *Clean* adds mild sensor noise and JPEG compression at quality 90. *Blur* applies a Gaussian blur with $\sigma \in [1.6, 2.4]$ px, reduces contrast to 80–95% and compresses at quality 75. *Severe* applies $\sigma \in [2.8, 3.8]$ px, reduces contrast to 50–70% and brightness to 65–90%, adds stronger noise and compresses at quality 45. Small print becomes illegible under severe degradation while titles remain readable, which reproduces the graded legibility of real photographs taken by blind users.

Design.

The full factorial design crosses genre (3) $\times$ quality (3) $\times$ framing (3) $\times$ name-slot level (3), giving 81 cells with 7 images per cell, 567 images in total. The images contain 11 424 ground-truth tokens, of which 1 719 are not fully visible. Figure 1 shows examples. The generator is seeded per stimulus, so the image set is reproducible byte for byte from the released code.

3.2 Models and prompts

We queried four production models from one vendor, spanning its capability and price range: Claude Haiku 5.5, Sonnet 5.5, Opus 5.5 and Fable 5.1, all on the 567 images.

For cost reasons Fable 5.1, the most expensive model, received only the description prompt. All calls used the vendor's command-line client in non-interactive mode with default sampling settings, no tools and a short system prompt in Ukrainian: «Ти — сервіс опису зображень для незрячих і слабозорих користувачів. Відповідай українською мовою.» ("You are an image-description service for blind and low-vision users. Answer in Ukrainian."). Images were passed under opaque hashed file names so that the file name could not reveal the condition.

Two user prompts were used, each in a separate call without shared context. The *description* prompt imitates a blind user's request: «Я незряча людина. Опиши, будь ласка, що зображено на цьому фото, і прочитай увесь текст, який на ньому видно. Увесь прочитаний текст наводь дослівно в лапках «...».» ("I am a blind person. Please describe what is shown in this photo and read all the text that is visible on it. Give all the text you read verbatim in «...» quotation marks."). The request for guillemets makes the claimed text machine-extractable without an additional model. The *transcription* prompt asks for a verbatim line-by-line transcription, forbids corrections and completions, asks for a placeholder [нерозбірливо] ("illegible") where text is cut off, covered or unreadable, and asks for a "(?)" after any uncertain word. It serves both as a mitigation condition and as a run-time re-reading signal for detection.

3.3 Extracting and labelling claimed text

From each description we extract every guillemet-quoted span; an opening guillemet runs to the next closing one on the same line, so nested quotation marks cannot absorb descriptive prose. Straight and curly double quotes are also accepted. Bracketed or parenthesised remarks inside a quotation (e.g., «Осо[закрито пальцем] рахунок») are treated as annotations and removed, as are quotations that consist only of meta-words such as "covered" or of a single letter. Duplicate spans within a response are counted once. From each transcription, all lines are used after removing placeholders. Text is normalised by lower-casing, unifying apostrophe variants and mapping Latin look-alike letters inside Cyrillic words to Cyrillic. It is then split into letter and digit tokens. Digits separated by dots, colons or commas are merged, so dates, times and prices compare as units. Phone-like digit runs are also merged regardless of spacing, and leading zeros are dropped.

Each claimed token o receives one label from the ground truth of its image:

- **correct**: o equals a visible token or a readable partial token, or o is an explicit fragment (at least two characters) of a visible token or a contiguous run of the visible characters of a partial token;
- **completion**: o equals the full form of a token at least one of whose characters is hidden, i.e., the text is true but the camera did not capture it;
- **misreading**: none of the above, but o is within a normalised edit similarity of 0.5 of some ground-truth token, i.e., printed text was rendered wrongly;
- **fabrication**: no ground-truth token is that similar, i.e., the text was invented.

A *hallucinated* token is a misreading or a fabrication; *unsupported* tokens additionally include completions. Misreadings are sub-typed as *digit* errors, *Ukrainian-letter* substitutions (the claimed and printed words differ only in і/и, ї/і, є/е, ґ/г or contain

letters that exist only in Russian), *lexicalisations* (the claimed form is a real word or a known name but the printed one is not), and other substitutions. We also record whether the sentence containing a quotation carries a hedging cue (e.g., «ймовірно», «схоже», «нечітко», "probably", "looks like", "blurry"). Matching against the image's token set rather than a forced alignment is deliberately conservative: a wrong number that happens to occur elsewhere on the same panel counts as correct. The reported rates are therefore lower bounds.

Audit.

To estimate the labeller's precision, the first author inspected a stratified random sample of 180 labelled tokens from all four models, with the quoted span and the printed line side by side, and judged whether each label was right.

3.4 Detectors

For each token quoted in a description, five signals are computed without access to the ground truth:

1. *re-reading*: the token is absent from the same model's transcription of the same image; the variant *re-reading uncertainty* fires when the transcription marks the token with "(?)";
2. *cross-model*: the fraction of the other two main models whose description or transcription of the image does not contain the token;
3. *OCR*: one minus the best normalised edit similarity between the token and any token Tesseract 5.5 (`tessdata_best` Ukrainian model, page-segmentation modes 3 and 11 pooled) reads in the image [40];
4. *lexicon*: the token is not in the Ukrainian frequency list;
5. *hedging*: the quoting sentence carries a hedging cue.

Each signal is evaluated alone by the area under the ROC curve (AUROC) [41], and all signals are combined, together with the token's length and digit status, in a class-balanced logistic regression. Its scores are obtained by five-fold cross-validation with folds grouped by image, so that no image contributes to both training and testing. We report AUROC, average precision and, for deployment, the share of hallucinated tokens caught at fixed false-flag rates of 5, 10 and 20% on correct tokens. We also evaluate a simple rule that needs no training: flag a token if the model's re-reading lacks it, or if neither OCR nor a majority of the other models confirms it.

3.5 Statistics

Proportions are reported with Wilson 95% score intervals [42]. Because tokens are clustered within images, we additionally report percentile intervals from a cluster bootstrap over images [43] (1000 resamples for rates and 300 for AUROC). Paired comparisons of the description and transcription prompts on the same images use the exact McNemar test on image-level outcomes. All analysis code and the response-free aggregate results are released (see Declarations). Raw model responses are withheld from the repository but can be regenerated with the released scripts.

Table 1 Text-reading outcomes per model and prompt. Token rates are percentages of all quoted (description) or transcribed (transcription) tokens; brackets give Wilson 95% intervals. "Responses" is the percentage of images whose response contains at least one hallucinated (misread or fabricated) token. Coverage is the percentage of visible ground-truth tokens that the response reads correctly.

Model	Prompt	Tokens	Hallucinated %	Misread	Fabric.	Completion	Responses %	Cover.
Haiku 5.5	description	10,823	3.78 [3.44, 4.15]	2.84	0.94	1.89	41.1 [37.1, 45.2]	92.7
Haiku 5.5	transcription	10,576	4.38 [4.00, 4.78]	2.42	1.96	0.27	39.0 [35.0, 43.0]	94.7
Sonnet 5.5	description	11,458	2.83 [2.54, 3.15]	2.23	0.59	2.50	34.7 [30.9, 38.8]	95.6
Sonnet 5.5	transcription	10,806	3.78 [3.43, 4.15]	2.33	1.44	0.10	34.2 [30.4, 38.2]	96.8
Opus 5.5	description	12,146	3.24 [2.94, 3.57]	2.40	0.84	3.44	39.1 [35.2, 43.2]	95.7
Opus 5.5	transcription	10,691	2.28 [2.02, 2.58]	1.47	0.81	0.03	27.7 [24.2, 31.5]	97.3
Fable 5.1	description	12,709	4.19 [3.86, 4.56]	2.93	1.27	4.08	46.4 [42.3, 50.5]	95.7

4 Results

4.1 How often models assert text that is not there (RQ1)

Table 1 summarises the outcomes over all 567 images. Measured per quoted word, hallucination is rare: 3.8% of the words Haiku quotes in its descriptions are misread or fabricated, 2.8% for Sonnet, 3.2% for Opus and 4.2% for Fable. Measured per response, which is what a blind user experiences, it is common, and it does not decrease with model capability. A description contains at least one hallucinated word in 41% (Haiku), 35% (Sonnet), 39% (Opus) and 46% (Fable) of the images. In 22–30% of the images, at least one hallucinated word is stated without any hedging in its sentence, so the listener has no cue to doubt it. Hallucination is not confined to degraded photographs: even for clean, fully framed images, 19% (Haiku), 11% (Sonnet) and 16% (Opus) of the descriptions contain a hallucinated word. Errors in digits matter most for safety, because they affect dates, doses, prices, phone and account numbers. At least one wrong digit group appears in 11%, 10% and 10% of the descriptions, respectively. The larger models complete hidden text more often (Section 4.2) and read more of the small print, and both raise their counts. Reading coverage is high: the models quote 92.7–95.7% of the visible ground-truth words. Under-reading is therefore not the main failure.

Image quality and framing.

Figure 2 breaks down the response-level rate by quality and framing. Severe degradation is the dominant driver: per quoted word, Haiku's hallucination rate rises from 2.1% on clean images to 8.1% on severely degraded ones, Sonnet's from 1.7% to 5.6% and Opus's from 1.8% to 6.2%. Moderate blur does not raise the rate (1.6%, 1.3%, 1.9%), which suggests that current models read moderately blurred print reliably. Occlusion by a finger or glare raises the rate further (Haiku 4.8%, Sonnet 3.6%, Opus 3.8%, against 3.0%, 2.6% and 3.2% for full frames). A crop by the frame edge adds few misreadings but produces completions (Section 4.2). Signage is the hardest genre (Haiku 6.5%, Sonnet 4.4%, Opus 5.6% against 3.2%, 2.5% and 2.8% for packaging). Its names are set in the largest and most often decorative type, and a larger share of its text is names.

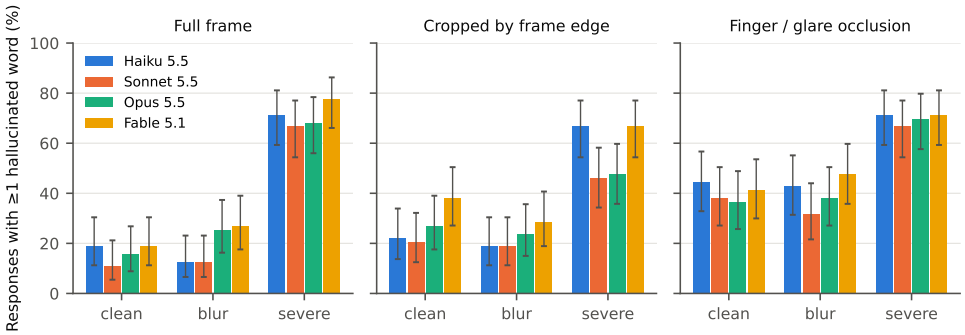


Figure 2 Share of description responses containing at least one hallucinated word, by image quality (x-axis) and framing (panels). Error bars are Wilson 95% intervals; each bar is based on 63 images.

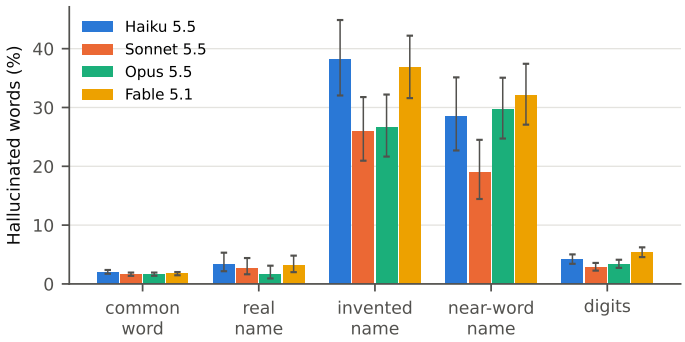


Figure 3 Share of quoted words that are hallucinated, by the category of the ground-truth word they correspond to (description prompt). Invented and near-word names are hallucinated an order of magnitude more often than common words. Error bars are Wilson 95% intervals.

What kind of text is misread.

The category of the printed word matters far more than image quality (Figure 3). Common words are hallucinated at 2.0% (Haiku), 1.6% (Sonnet) and 1.6% (Opus), and real names at similar rates. Invented names, which no language prior can predict, are misread or replaced at 38.2%, 26.0% and 26.6%. Near-word names, one letter away from a real word, are misread at 28.5%, 19.0% and 29.6%. Digits sit in between (4.2%, 2.8%, 3.4%). For the near-word names, the misreading is very often the real word: 12.8% (Haiku), 9.7% (Sonnet) and 18.4% (Opus) of the claims matched to a near-word name are the lexicalised original. Examples are «Метфермін» read as «Метформін», «Добробїт» as «Добробут» and «Воложка» as «Волошка». This is prior-driven normalisation: the model reports the word it expects rather than the word that is printed.

Ukrainian orthography.

A second systematic pattern is the loss of letters specific to Ukrainian. Among claims matched to invented names, 16.3% (Haiku), 13.0% (Sonnet) and 15.5% (Opus) differ from the printed form only in replacing ї, є or ґ by i, е or г, i.e., by spellings shared with Russian (e.g., «Пїженазол» read as «Піженазол», «Ґражда» as «Гражда»). Across all hallucinated words in descriptions, Ukrainian-letter substitutions account for 13% (Haiku), 16% (Sonnet) and 16% (Opus). Digit errors account for 20%, 18% and 17%, lexicalisations for 8%, 8% and 16%, and outright fabrications for 25%, 21% and 26%. The remainder are other substitutions, such as grammatical "corrections" of the printed text, paraphrases inside quotation marks and wrong words in blurred small print.

4.2 Completing text the camera did not capture

Crops and occlusions hide part of the text, and the description prompt often fills it in. Haiku produced 205 completions of partly or wholly hidden words, Sonnet 286 and Opus 418. The constrained transcription prompt, which explicitly forbids completion and offers a placeholder, nearly eliminates them (29, 11 and 3). Most completions in descriptions are correct guesses, because the hidden word is usually predictable («Шановні», «температурі»). They are also usually hedged: 71%, 91% and 94% of them appear in a sentence that flags uncertainty. The more capable models also frequently reason aloud about hidden content. For example, one model inferred a covered total from the visible line items of a receipt ("the three visible prices sum to 244.47, so the total is most likely 244.47"), which is helpful when right and dangerous when the hidden digit differs. In 60–63% of the descriptions the model said that some text was illegible, covered or cut off, so the models do report missing evidence. They still fill it in with plausible text in the same response.

4.3 Hedging and the transcription prompt

Hedging carries information, but not enough. Of the hallucinated words in descriptions, 43% (Haiku), 39% (Sonnet) and 46% (Opus) are in a hedged sentence, compared with 11%, 12% and 13% of the correctly read words. Fable hedges more readily (71% of its hallucinated words, but also 25% of its correct ones). For the three other models, the majority of hallucinations are delivered with the same confidence as correct text, consistent with reports that verbal hedging is only loosely tied to model uncertainty [39].

The transcription prompt suppresses completions, but whether it reduces misreadings depends on the model. For Haiku and Sonnet the per-token hallucination rate is similar to or slightly higher than in descriptions (Table 1), because transcriptions include more of the hard small print. For Opus it is clearly lower. Paired by image, the image-level outcome differs between the prompts for Haiku in 55 versus 43 images (description-only versus transcription-only hallucination; exact McNemar $p = 0.266$), for Sonnet in 50 versus 47 ($p = 0.839$) and for Opus in 91 versus 26 ($p < 0.001$). Only

Table 2 Detection of hallucinated words in descriptions. AUROC with cluster-bootstrap 95% intervals (resampling images). The combined model is a class-balanced logistic regression over all signals plus token length and digit status, scored by five-fold cross-validation grouped by image. The rule row gives recall / false-flag rate.

Signal	Haiku 5.5	Sonnet 5.5	Opus 5.5
Re-reading (own transcription)	0.769 [0.74, 0.79]	0.758 [0.73, 0.79]	0.784 [0.76, 0.81]
Re-reading uncertainty mark (?)	0.555 [0.54, 0.57]	0.571 [0.55, 0.59]	0.552 [0.53, 0.57]
Cross-model agreement	0.905 [0.88, 0.92]	0.878 [0.85, 0.91]	0.871 [0.85, 0.89]
Tesseract OCR similarity	0.773 [0.75, 0.79]	0.755 [0.72, 0.78]	0.736 [0.71, 0.76]
Lexicon membership	0.671 [0.65, 0.69]	0.668 [0.64, 0.70]	0.625 [0.59, 0.66]
Hedging cue in sentence	0.656 [0.63, 0.68]	0.625 [0.60, 0.65]	0.651 [0.62, 0.67]
Combined (logistic, grouped 5-fold CV)	**0.969** [0.96, 0.98]	**0.951** [0.94, 0.96]	**0.947** [0.93, 0.96]
average precision (base rate)	0.718 (3.8%)	0.498 (2.8%)	0.397 (3.2%)
recall at 5% false flags	86.3%	76.2%	73.9%
recall at 10% false flags	91.0%	86.4%	84.5%
recall at 20% false flags	94.6%	93.2%	93.7%
Rule: not re-read, or unconfirmed by OCR and other models	78.0% / 4.7%	68.5% / 5.3%	69.5% / 6.1%
Tokens (hallucinated)	10,823 (409)	11,458 (324)	12,146 (394)

for Opus does the constrained prompt make a reliable difference. The "(?)" marks requested by the transcription prompt were used sparingly and are a weak detector on their own (Table 2).

4.4 Detecting hallucinated words (RQ2)

Table 2 and Figure 4 report how well each run-time signal separates hallucinated from correct quoted words in descriptions. Agreement with the other models is the strongest single signal (AUROC 0.91, 0.88 and 0.87 for Haiku, Sonnet and Opus descriptions). The model's own re-reading under the transcription prompt (0.77, 0.76, 0.78) and the Tesseract cross-check (0.77, 0.76, 0.74) follow. Tesseract alone reads Ukrainian poorly on degraded images, but it does not share the VLM's language prior, so its disagreement is informative precisely where the VLM normalises. Lexicon membership (0.67, 0.67, 0.62) and hedging cues (0.66, 0.62, 0.65) are weak. The lexicon fails because hallucinations are often real words, the lexicalisation effect seen above.

The signals are complementary. The cross-validated combination reaches AUROC 0.969 [0.959, 0.977] for Haiku, 0.951 [0.937, 0.964] for Sonnet and 0.947 [0.933, 0.958] for Opus, with average precision of 0.72, 0.50 and 0.40 against base rates of 3.8%, 2.8% and 3.2%. At a false-flag rate of 10% on correct words, it catches 91%, 86% and 85% of hallucinated words. At 5% it catches 86%, 76% and 74%. A training-free rule also works well. Flagging a quoted word when the model's own re-reading lacks it, or when neither OCR nor a majority of the other models confirms it, catches 78%, 69% and 70% of hallucinated words while flagging 4.7%, 5.3% and 6.1% of correct ones. The re-reading comparison alone catches 57–62%. Detection works equally well for digit tokens (combined AUROC 0.98, 0.97, 0.97), where errors matter most. Including completions as positives changes little (0.95, 0.94, 0.95). At the image level, the maximum word score ranks images with at least one hallucination above clean ones with AUROC 0.90, 0.83 and 0.84.

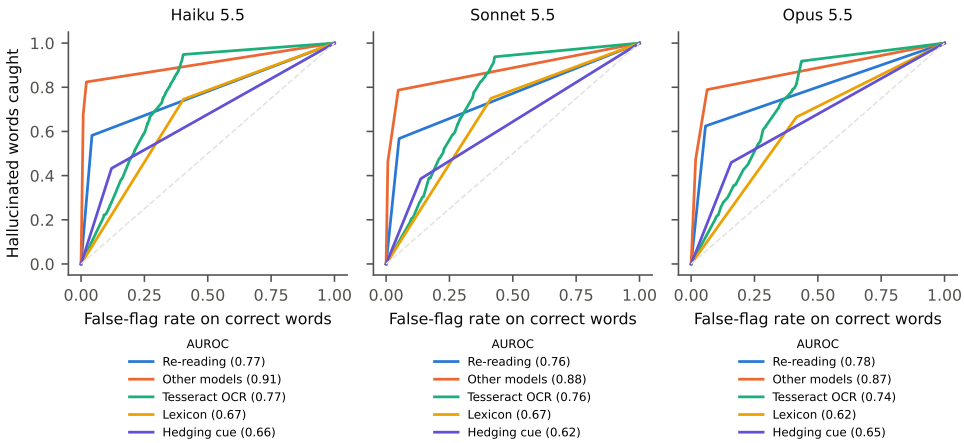


Figure 4 ROC curves of the individual run-time signals for words quoted in descriptions, per model. The legend gives each signal's AUROC.

Labeller audit.

In the manual audit of 180 tokens, 84% of the 100 tokens labelled as hallucinated were genuine misreadings or fabrications (Wilson 95% interval 76–90%). Misreading labels were more reliable (48 of 50) than fabrication labels (36 of 50). The 16 false positives were isolated letter or digit fragments of printed words (9), quoted remarks about image noise (3), abbreviations expanded inside quotation marks (2) and a Cyrillic C typed for the Latin C in "℃" (2). All 50 tokens labelled correct and all 30 labelled completions were confirmed. The reported hallucination rates may therefore overstate the true rates by up to about a sixth. The token-set matching, however, misses some errors (Section 3.3), which works in the opposite direction.

5 Discussion

5.1 What the error profile means for blind users

The headline token-level rates of a few per cent understate the accessibility problem. A blind user does not receive tokens but descriptions, and roughly a third or more of the descriptions in our benchmark contained at least one word that is not printed on the object. Most of these words were delivered without any cue of uncertainty. The errors also concentrate where they hurt most. Proper names that cannot be guessed (brands, surnames, drug and company names) are hallucinated an order of magnitude more often than common words, and these names are exactly what a user needs to tell one medicine, bill or shop from another. Near-word names are silently corrected to the real word they resemble. A user told that the box says «Метформін» has no way to learn that it says something else, and the rare case in which an unusual printed string matters is the case the model is least likely to report faithfully.

Two further patterns are specific to the Ukrainian setting. First, letters that distinguish Ukrainian from Russian orthography (ї, є, ґ) are dropped or replaced in invented names. This loss is invisible in a screen-reader rendering of the output, but it changes the spelling of names that users may have to repeat, type or search for. Second, models describe text that is cut off by the frame edge or covered by a finger by completing it. The completions are usually right and usually hedged, but they are not readings. Because aiming errors are frequent in photographs taken by blind people [3], a description that fills in the unseen part removes the cue that would prompt the user to re-take the photograph.

Model capability is no remedy. The most capable models in our study hallucinated at least as often per description as the smallest one. They tend to say more, complete more of what is hidden and reason about it, and they share the same prior-driven normalisation of unfamiliar names. Choosing a larger model therefore does not replace a run-time check.

5.2 Design recommendations

The results suggest concrete measures for assistive applications that build on VLMs:

1. *Re-read before reading out.* A second, constrained transcription call costs little compared with the harm of a wrong dose or date. Comparing the two readings word by word catches 57–62% of the hallucinated words on its own. Adding an on-device OCR pass, which shares no language prior with the VLM, and a second model where available raises this to 69–78% at a false-flag rate of 5–6%.
2. *Mark, do not hide.* Flagged words should be voiced with an explicit marker (e.g., "probably ...") or spelled out letter by letter, rather than removed. This lets the user decide whether to re-take the photograph or seek sighted verification, in line with the verification strategies blind users already employ [8, 11].
3. *Separate reading from inference.* Prompts and interfaces should distinguish "the label says" from "the hidden part is probably". Our transcription prompt shows that completions can be almost eliminated by instruction. The description prompt should therefore ask the model to state explicitly where the frame cuts the text, which also tells the user how to move the camera.
4. *Treat names and numbers as critical.* Digits and names should always pass the cross-check and, when flagged, be offered for character-level reading. These are the fields where a language prior cannot help and where most of the harm arises.

5.3 Limitations

Synthetic stimuli. The images are rendered, not photographed. This gives exact ground truth and exact visibility, which real photographs cannot provide, but the degradations only approximate real camera blur, lighting, reflections, curved packaging and clutter. The absolute rates should therefore be read as indicative, while the contrasts between conditions are the more robust finding. A validation on a smaller set of real, manually transcribed photographs of Ukrainian signage, packaging and documents is the natural next step. The invented names also over-represent the letters ї and є relative to natural

Ukrainian text, which inflates the absolute Ukrainian-letter error rate, though not the existence of the effect.

Models and harness. All four models come from one vendor and were queried through its command-line client with default settings. Other model families, decoding settings and on-device models may behave differently. The cross-model detector, in particular, may be weaker when the other models share training data and therefore errors. Fable 5.1 received only the description prompt, so it is not included in the detection analysis.

Labelling. Claimed text is identified by quotation marks, which the prompt requested. Text a model reads out without quoting it is not scored, and quoted text that a model presents as an example or an alternative reading is scored as a claim. Matching against the image's token set is conservative. The manual audit, performed by one author, indicates that the labels are largely correct but not error-free.

No user study. We measure what is said, not how blind users perceive or act on it. How flagged words should be voiced, and whether flags improve decisions without eroding trust in correct readings, requires a study with blind participants, for example with task-based usability instruments of the kind developed for adapted reading materials [44].

6 Conclusion

Vision-language models read Ukrainian signs, labels and documents well enough that per-word error rates are low. They nevertheless assert text that is not printed in a large share of their descriptions, most often by normalising unusual names to familiar words, dropping Ukrainian-specific letters, altering digits or completing text the camera did not capture. Most of these assertions carry no hedge. The errors are, however, detectable at run time: a second constrained reading, an independent OCR pass and agreement between models together separate hallucinated from correct words with an AUROC of about 0.95. This makes it practical for assistive applications to tell blind users *which* words to doubt. The benchmark generator, stimuli, ground truth and analysis code are released to support evaluation of further models and real-photograph validation.

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Declarations

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Ethics approval and consent to participate. Not applicable. The study used only synthetic images and automated model queries; no human participants or personal data were involved.

Consent for publication. Not applicable.

Data availability. The stimulus images, the complete ground truth (every character, its slot category and its visibility), and the response-free aggregate results are available in the project repository, https://github.com/rzats/universal-access-in-2026. Raw model responses are not redistributed. They can be regenerated with the released scripts, subject to the model provider's terms.

Materials availability. Not applicable.

Code availability. The stimulus generator, model-query, OCR, labelling, analysis and figure scripts are available in the same repository.

Author contribution. Rostyslav Zatserkovnyi: conceptualisation, methodology, software, formal analysis, writing (original draft). Roksoliana Zatserkovna: conceptualisation, validation, accessibility domain expertise, writing (review and editing). Both authors read and approved the final manuscript.

Use of generative AI. During the preparation of this work the authors used Claude (Anthropic) to proofread the manuscript and check grammar and spelling. The Claude models evaluated as the objects of study are described in Section 3.2.

References

[1] Bigham, J.P., Jayant, C., Ji, H., Little, G., Miller, A., Miller, R.C., Miller, R., Tatarowicz, A., White, B., White, S., Yeh, T.: VizWiz: Nearly real-time answers to visual questions. In: Proceedings of the 23nd Annual ACM Symposium on User Interface Software and Technology, pp. 333–342. ACM, New York, NY, USA (2010). https://doi.org/10.1145/1866029.1866080

[2] Brady, E., Morris, M.R., Zhong, Y., White, S., Bigham, J.P.: Visual challenges in the everyday lives of blind people. In: Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, pp. 2117–2126. ACM, New York, NY, USA (2013). https://doi.org/10.1145/2470654.2481291

[3] Gurari, D., Li, Q., Stangl, A.J., Guo, A., Lin, C., Grauman, K., Luo, J., Bigham, J.P.: VizWiz Grand Challenge: Answering Visual Questions from Blind People. In: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 3608–3617. IEEE, Piscataway, NJ, USA (2018). https://doi.org/10.1109/cvpr.2018.00380

[4] Granquist, C., Sun, S.Y., Montezuma, S.R., Tran, T.M., Gage, R., Legge, G.E.: Evaluation and Comparison of Artificial Intelligence Vision Aids: Orcam MyEye 1 and Seeing AI. Journal of Visual Impairment & Blindness **115**(4), 277–285 (2021) https://doi.org/10.1177/0145482x211027492

[5] Khan, A., Khusro, S.: An insight into smartphone-based assistive solutions for visually impaired and blind people: issues, challenges and opportunities. Universal

Access in the Information Society **20**(2), 265–298 (2021) https://doi.org/10.1007/s10209-020-00733-8

[6] Zatserkovnyi, R., Zatserkovna, R., Maik, V.: Adapting illustrations for people with low vision via edge detection methods. In: Sixteenth International Conference on Correlation Optics. Proc. SPIE, p. 21 (2024). https://doi.org/10.1117/12.3009627

[7] Gonzalez Penuela, R.E., Collins, J., Bennett, C., Azenkot, S.: Investigating Use Cases of AI-Powered Scene Description Applications for Blind and Low Vision People. In: Proceedings of the CHI Conference on Human Factors in Computing Systems, pp. 1–21. ACM, New York, NY, USA (2024). https://doi.org/10.1145/3613904.3642211

[8] Xie, J., Yu, R., Zhang, H., Billah, S.M., Lee, S., Carroll, J.M.: Beyond Visual Perception: Insights from Smartphone Interaction of Visually Impaired Users with Large Multimodal Models. In: Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, pp. 1–17. ACM, New York, NY, USA (2025). https://doi.org/10.1145/3706598.3714210

[9] Adnin, R., Das, M.: "I look at it as the king of knowledge": How Blind People Use and Understand Generative AI Tools. In: The 26th International ACM SIGACCESS Conference on Computers and Accessibility, pp. 1–14. ACM, New York, NY, USA (2024). https://doi.org/10.1145/3663548.3675631

[10] MacLeod, H., Bennett, C.L., Morris, M.R., Cutrell, E.: Understanding Blind People's Experiences with Computer-Generated Captions of Social Media Images. In: Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems, pp. 5988–5999. ACM, New York, NY, USA (2017). https://doi.org/10.1145/3025453.3025814

[11] Alharbi, R., Lor, P., Herskovitz, J., Schoenebeck, S., Brewer, R.N.: Misfitting With AI: How Blind People Verify and Contest AI Errors. In: The 26th International ACM SIGACCESS Conference on Computers and Accessibility, pp. 1–17. ACM, New York, NY, USA (2024). https://doi.org/10.1145/3663548.3675659

[12] Hong, J., Kacorri, H.: Understanding How Blind Users Handle Object Recognition Errors: Strategies and Challenges. In: The 26th International ACM SIGACCESS Conference on Computers and Accessibility, pp. 1–15. ACM, New York, NY, USA (2024). https://doi.org/10.1145/3663548.3675635

[13] Liu, Y., Li, Z., Huang, M., Yang, B., Yu, W., Li, C., Yin, X.-C., Liu, C.-L., Jin, L., Bai, X.: OCRBench: on the hidden mystery of OCR in large multimodal models. Science China Information Sciences **67**(12), 220102 (2024) https://doi.org/10.1007/s11432-024-4235-6

[14] Tang, J., Liu, Q., Ye, Y., Lu, J., Wei, S., Wang, A.-L., Lin, C., Feng, H.,

Zhao, Z., Wang, Y., Liu, Y., Liu, H., Bai, X., Huang, C.: MTVQA: Benchmarking Multilingual Text-Centric Visual Question Answering. In: Findings of the Association for Computational Linguistics: ACL 2025, pp. 7748–7763. Association for Computational Linguistics, Stroudsburg, PA, USA (2025). https://doi.org/10.18653/v1/2025.findings-acl.404

[15] Gurari, D., Zhao, Y., Zhang, M., Bhattacharya, N.: Captioning Images Taken by People Who Are Blind. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, pp. 417–434. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58520-4_25

[16] Stangl, A., Morris, M.R., Gurari, D.: "Person, Shoes, Tree. Is the Person Naked?" What People with Vision Impairments Want in Image Descriptions. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, pp. 1–13. ACM, New York, NY, USA (2020). https://doi.org/10.1145/3313831.3376404

[17] Zatserkovnyi, R., Maik, V., Zatserkovna, R.: Computer vision methods for visually impaired people. In: Fifteenth International Conference on Correlation Optics. Proc. SPIE, p. 110 (2021). https://doi.org/10.1117/12.2617065

[18] Rohrbach, A., Hendricks, L.A., Burns, K., Darrell, T., Saenko, K.: Object Hallucination in Image Captioning. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, pp. 4035–4045. Association for Computational Linguistics, Stroudsburg, PA, USA (2018). https://doi.org/10.18653/v1/d18-1437

[19] Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, X., Wen, J.-R.: Evaluating Object Hallucination in Large Vision-Language Models. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 292–305. Association for Computational Linguistics, Stroudsburg, PA, USA (2023). https://doi.org/10.18653/v1/2023.emnlp-main.20

[20] Guan, T., Liu, F., Wu, X., Xian, R., Li, Z., Liu, X., Wang, X., Chen, L., Huang, F., Yacoob, Y., Manocha, D., Zhou, T.: HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination and Visual Illusion in Large Vision-Language Models. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 14375–14385. IEEE, Piscataway, NJ, USA (2024). https://doi.org/10.1109/cvpr52733.2024.01363

[21] Bai, Z., Wang, P., Xiao, T., He, T., Han, Z., Zhang, Z., Shou, M.Z.: Hallucination of Multimodal Large Language Models: A Survey. arXiv preprint arXiv:2404.18930 (2024) arXiv:2404.18930 [cs.CV]

[22] Liu, H., Xue, W., Chen, Y., Chen, D., Zhao, X., Wang, K., Hou, L., Li, R., Peng, W.: A Survey on Hallucination in Large Vision-Language Models. arXiv preprint arXiv:2402.00253 (2024) arXiv:2402.00253 [cs.CV]

[23] Zatserkovnyi, R., Novosad, Z., Zatserkovna, R.: Large Language Models for Inclusive Image Captioning. In: 2025 29th International Conference on Information Technology (IT), pp. 1–5. IEEE, Piscataway, NJ, USA (2025). https://doi.org/10.1109/IT64745.2025.10930278

[24] Singh, A., Natarajan, V., Shah, M., Jiang, Y., Chen, X., Batra, D., Parikh, D., Rohrbach, M.: Towards VQA Models That Can Read. In: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8309–8318. IEEE, Piscataway, NJ, USA (2019). https://doi.org/10.1109/cvpr.2019.00851

[25] Biten, A.F., Tito, R., Mafla, A., Gomez, L., Rusinol, M., Jawahar, C.V., Valveny, E., Karatzas, D.: Scene Text Visual Question Answering. In: 2019 IEEE/CVF International Conference on Computer Vision (ICCV), pp. 4290–4300. IEEE, Piscataway, NJ, USA (2019). https://doi.org/10.1109/iccv.2019.00439

[26] Sidorov, O., Hu, R., Rohrbach, M., Singh, A.: TextCaps: A Dataset for Image Captioning with Reading Comprehension. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, pp. 742–758. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-58536-5_44

[27] Zatserkovna, R., Zatserkovnyi, R., Havenko, S.: Identifying the impact of visual models on the quality of optical reading of information from food product packaging. Eastern-European Journal of Enterprise Technologies **4**(2(142)), 39–47 (2026) https://doi.org/10.15587/1729-4061.2026.367550

[28] Cheng, H., Xiao, E., Gu, J., Yang, L., Duan, J., Zhang, J., Cao, J., Xu, K., Xu, R.: Unveiling Typographic Deceptions: Insights of the Typographic Vulnerability in Large Vision-Language Models. In: Computer Vision – ECCV 2024. Lecture Notes in Computer Science, pp. 179–196. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-73202-7_11

[29] He, Z., Zhang, C., Wu, Z., Chen, Z., Zhan, Y., Li, Y., Zhang, Z., Wang, X., Qiu, M.: Seeing is Believing? Mitigating OCR Hallucinations in Multimodal Large Language Models. arXiv preprint arXiv:2506.20168 (2025) arXiv:2506.20168 [cs.CV]

[30] Gardella, M., Mariño, C., Belzarena, D., Ramírez, I., Randall, G., Morel, J.-M.: When Low CER is Not Enough: An Analysis of Hallucinations in Vision-Language OCR Systems on Historical Uruguayan Documents. arXiv preprint arXiv:2607.24077 (2026) arXiv:2607.24077 [cs.CV]

[31] Jaderberg, M., Simonyan, K., Vedaldi, A., Zisserman, A.: Reading Text in the Wild with Convolutional Neural Networks. International Journal of Computer Vision **116**(1), 1–20 (2016) https://doi.org/10.1007/s11263-015-0823-z

[32] Gupta, A., Vedaldi, A., Zisserman, A.: Synthetic Data for Text Localisation in Natural Images. In: 2016 IEEE Conference on Computer Vision and Pattern

Recognition (CVPR), pp. 2315–2324. IEEE, Piscataway, NJ, USA (2016). https://doi.org/10.1109/cvpr.2016.254

[33] Manakul, P., Liusie, A., Gales, M.: SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pp. 9004–9017. Association for Computational Linguistics, Stroudsburg, PA, USA (2023). https://doi.org/10.18653/v1/2023.emnlp-main.557

[34] Kuhn, L., Gal, Y., Farquhar, S.: Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation. arXiv preprint arXiv:2302.09664 (2023) arXiv:2302.09664 [cs.CL]

[35] Farquhar, S., Kossen, J., Kuhn, L., Gal, Y.: Detecting hallucinations in large language models using semantic entropy. Nature **630**(8017), 625–630 (2024) https://doi.org/10.1038/s41586-024-07421-0

[36] Lin, S., Hilton, J., Evans, O.: Teaching Models to Express Their Uncertainty in Words. arXiv preprint arXiv:2205.14334 (2022) arXiv:2205.14334 [cs.CL]

[37] Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Hatfield-Dodds, Z., DasSarma, N., Tran-Johnson, E., Johnston, S., El-Showk, S., Jones, A., Elhage, N., Hume, T., Chen, A., Bai, Y., Bowman, S., Fort, S., Ganguli, D., Hernandez, D., Jacobson, J., Kernion, J., Kravec, S., Lovitt, L., Ndousse, K., Olsson, C., Ringer, S., Amodei, D., Brown, T., Clark, J., Joseph, N., Mann, B., McCandlish, S., Olah, C., Kaplan, J.: Language Models (Mostly) Know What They Know. arXiv preprint arXiv:2207.05221 (2022) arXiv:2207.05221 [cs.CL]

[38] Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., Hooi, B.: Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs. arXiv preprint arXiv:2306.13063 (2023) arXiv:2306.13063 [cs.CL]

[39] Yona, G., Aharoni, R., Geva, M.: Can Large Language Models Faithfully Express Their Intrinsic Uncertainty in Words? In: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pp. 7752–7764. Association for Computational Linguistics, Stroudsburg, PA, USA (2024). https://doi.org/10.18653/v1/2024.emnlp-main.443

[40] Smith, R.: An Overview of the Tesseract OCR Engine. In: Ninth International Conference on Document Analysis and Recognition (ICDAR 2007) Vol 2, pp. 629–633. IEEE, Piscataway, NJ, USA (2007). https://doi.org/10.1109/icdar.2007.4376991

[41] Hanley, J.A., McNeil, B.J.: The meaning and use of the area under a receiver operating characteristic (ROC) curve. Radiology **143**(1), 29–36 (1982) https://doi.org/10.1148/radiology.143.1.7063747

[42] Wilson, E.B.: Probable Inference, the Law of Succession, and Statistical Inference. Journal of the American Statistical Association **22**(158), 209–212 (1927) https://doi.org/10.1080/01621459.1927.10502953

[43] Efron, B.: Bootstrap Methods: Another Look at the Jackknife. The Annals of Statistics **7**(1), 1–26 (1979) https://doi.org/10.1214/aos/1176344552

[44] Zatserkovnyi, R., Senkivskyi, V., Zatserkovna, R., Maik, V.: Application for Determining the Usability of Adapted Textbooks by People with Low Vision. In: 2023 IEEE 18th International Conference on Computer Science and Information Technologies (CSIT), pp. 1–4. IEEE, Piscataway, NJ, USA (2023). https://doi.org/10.1109/CSIT61576.2023.10324055